

LG-22-120

December 13, 2022

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket No. 50-352

Subject: LER 2022-001-00 Unit 1 HPCI Inoperable Due to Inadvertent Isolation Signal

In accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), Limerick Generating Station hereby submits the enclosed Licensee Event Report.

There are no commitments contained in this letter.

If you have any questions, please contact Jordan Rajan at (610) 718-2513.

Respectfully,



Michael Gillin
Vice President – Limerick Generating Station
Constellation Generation Company, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Limerick Generating Station Unit 1

2. Docket Number

05000

352

3. Page

1 OF 3

4. Title

Unit 1 HPCI Inoperable Due to Inadvertent Isolation Signal

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
10	14	22	2022	- 001 -	00	12	13	22	Facility Name	Docket Number
									Facility Name	Docket Number
										05000
										05000

9. Operating Mode

OPCON 1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER
Licensee Contact

Jordan Rajan

Phone Number (Include area code)

610-718-2513

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	BJ	CON	I215	Yes					

14. Supplemental Report Expected


No



Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On October 14, 2022, during planned testing of the High Pressure Coolant Injection (HPCI) System, a spurious HPCI Division 2 isolation occurred due to a fault in an electrical connector associated with the HPCI Division 2 isolation reset circuit. The affected electrical connector carries signals for the control circuits of the outboard HPCI Turbine Exhaust Line Vacuum Breaker isolation valve and HPCI Test Line Flush isolation valve and the HPCI Division 2 isolation reset circuit. Closing the outboard HPCI Turbine Exhaust Line Vacuum Breaker isolation valve energized degraded pins within the electrical connector resulting in a fault that initiated the HPCI Division 2 isolation signal. The defective electrical connector was removed, and the affected cable spliced in accordance with an approved design change. This event is being reported as a condition that, at the time of discovery, could have prevented the fulfillment of the safety function of a system needed to mitigate the consequences of an accident. An eight (8) hour ENS notification was completed on October 14, 2022, at 18:03 hours.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generating Station Unit 1	05000-352	2022	001	00

NARRATIVE**Unit Condition**

Prior to the Event Unit 1 was in Operational Condition (OPCON) 1 (Power Operation) at approximately 100% power. There were no structures, systems, or components out of service that contributed to this event.

Description of the Event

On October 14, 2022, a Unit 1 High Pressure Coolant Injection (HPCI) System [BJ] surveillance test was conducted on the HPCI Turbine Exhaust Vacuum Breakers. The purpose of this test is to ensure that the HPCI turbine exhaust line vacuum breakers function properly in both the forward and reverse flow directions. The HPCI turbine exhaust line vacuum breaker isolation valves, HV-055-1F093 and HV-055-1F095 must be closed to establish the required alignment to test the vacuum breaker check valves. The HPCI System is inoperable, but available in this test alignment. At 09:15 the HPCI turbine exhaust line vacuum breaker isolation valves were closed to establish the required alignment to perform the surveillance test. At 09:23, a Unit 1 Division 2 Safeguard Battery ground alarm was received, and all DC related testing was stopped to investigate the alarm. At 10:07, a HPCI Division 2 isolation signal was received accompanied by auto closure of the outboard HPCI Steam line isolation valve and the inboard HPCI Pump Suppression Pool suction valve. An eight (8) hour ENS notification (56163) was completed on October 14, 2022, at 18:03 hours.

Troubleshooting and failure analysis of the removed electrical connector concluded that degradation of several of the pin connections within the affected electrical connector resulted in a fault that propagated to the HPCI Division 2 isolation reset circuit. The fault within the electrical connector was initiated when the degraded pin connections were energized by closure of the outboard HPCI turbine exhaust line vacuum breaker isolation valve, HV-055-1F093, for the planned surveillance test. The defective electrical connector was removed, and the affected cable spliced in accordance with an approved design change. The HPCI system was returned to service on October 17, 2022, at 21:15.

Analysis of the Event

The inadvertent HPCI isolation that occurred during a planned surveillance test was attributed to a degraded electrical connector that was energized when closing one of the motor operated valves required to establish the alignment for the surveillance test.

Failure analysis of the electrical connector identified that a fault propagated to the HPCI Division 2 isolation reset circuit when the outboard HPCI turbine exhaust vacuum breaker isolation valve, HV-055-1F093, was closed to establish the required alignment to perform the planned surveillance test. Additionally, due to the degradation on other pin connectors of the affected electrical connector, the failure analysis could not conclusively eliminate the possibility of a fault occurring through a different mechanism.

With the unplanned loss of a single train system this condition is being reported per 10 CFR 50.73(a)(2)(v)(D), any condition that could have prevented the fulfillment of the safety function of structures or systems required to mitigate the consequences of an accident.

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Limerick Generating Station Unit 1	05000-352	YEAR 2022	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE**Safety Consequences**

There were no actual safety consequences associated with this event. The potential safety consequences of this event were minimal. The HPCI System was removed from service in accordance with Technical Specifications prior to the isolation. The Reactor Core Isolation Cooling (RCIC) System, Automatic Depressurization System (ADS), Residual Heat Removal (RHR) System, and Core Spray (CS) System remained operable during the period that HPCI was inoperable.

Corrective Actions Completed

The degraded electrical connector was removed, and the affected cable spliced in accordance with an approved design change.

Corrective Actions Planned

Evaluate the need to inspect a representative sample of similar electrical connectors in the Auxiliary Equipment Room cabinets.

Previous Similar Occurrences

There have been two recent similar events at Limerick involving degraded electrical connectors. Both events involved electrical connectors that affected motor operated valve control circuits for RCIC System valves. In January 2016, a Unit 2 RCIC Pump Minimum Flow Bypass Valve exhibited erratic performance during a RCIC pump surveillance test. The affected electrical connector was inspected, and several pins were cleaned to restore proper operation of the circuit. In November 2019, when cycling the RCIC Steam Line Warm-up Line isolation valve for in-service testing, a Unit 1 Division 1 Safeguard Battery Ground alarm was received. Two shorted conductors in the affected electrical connector were spared and replaced with two good spare conductors to restore the circuits to the as-designed configuration.

Component Data

System: BJ High Pressure Coolant Injection
Component: CON (Connector)
Manufacturer: I215 (Cannon ITT)